SMART INDIA HACKATHON 2025

Problem Statement ID - 25184

Problem Statement Title - Secure closed group communication platform over exisiting public mobile communication networks for defence personnal and families.

Theme - Blockchain & Cybersecurity

PS Category - Software

Team ID - 72808

Team Name - INNOVISION

RAKSHA-SETU

A SECURE COMMUNICATION ECOSYSTEM FOR THE MODERN MILITARY COMMUNITY

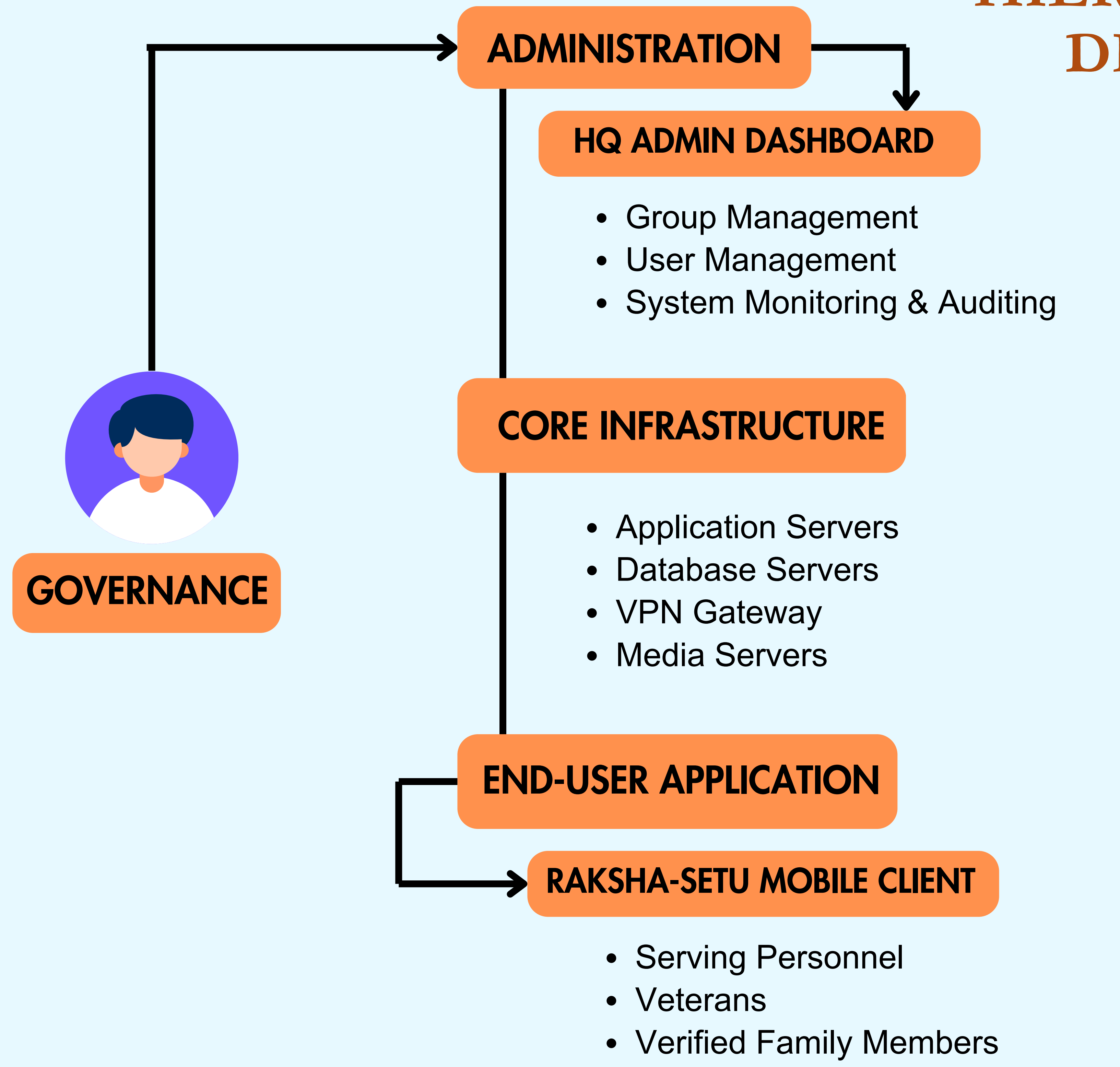
PROBLEM OVERVIEW

- Defence personnel using commercial apps risk data leaks and espionage.
- Foreign-hosted platforms expose sensitive defence communication.
- No sovereign, secure app exists for defence-exclusive communication.
- Commercial apps lack strict access control and data leak prevention.
- Sensitive family and operational details remain vulnerable on public platforms.

PROBLEM SOLUTION

- **HQ-Verified Closed Groups –** Only verified serving personnel, veterans, and registered families can access.
- **Mandatory VPN Encapsulation –** All communication routed through a defence-controlled VPN gateway.
- **End-to-End Encryption –** Messages, calls, and media remain fully encrypted.
- **Data Leak Prevention (DLP) –** Prevents screenshots, screen recording, copy-paste, and message forwarding.
- **On-Premise Hosting in India –** Ensures data sovereignty under military oversight.
- **Multi-modal Communication –** Secure text, multimedia sharing, and voice/video calling.

HIERARCHICAL DIAGRAM



TECHNICAL APPROACH

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METHODOLOGY

Gather defence-specific needs, security requirements, and user scenarios.
Requirement Analysis

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Define secure architecture with VPN, E2EE, and sovereign hosting.
System Architecture Design

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Deploy on-premise servers, VPN gateways, and databases in India.
Infrastructure Setup

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Build APIs, user authentication, and HQ admin dashboard.
Backend Development

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Create Android/iOS apps with onboarding, messaging, and group features.
Mobile App Development

→

Implement Signal Protocol and WireGuard for secure communication.
Encryption & VPN Integration

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Add secure voice/video calling and encrypted media sharing.
Media & Call Features

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Apply DLP, device health checks, and penetration testing.
Security Hardening

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Test with selected defence personnel and families for real-world validation.
Pilot Deployment

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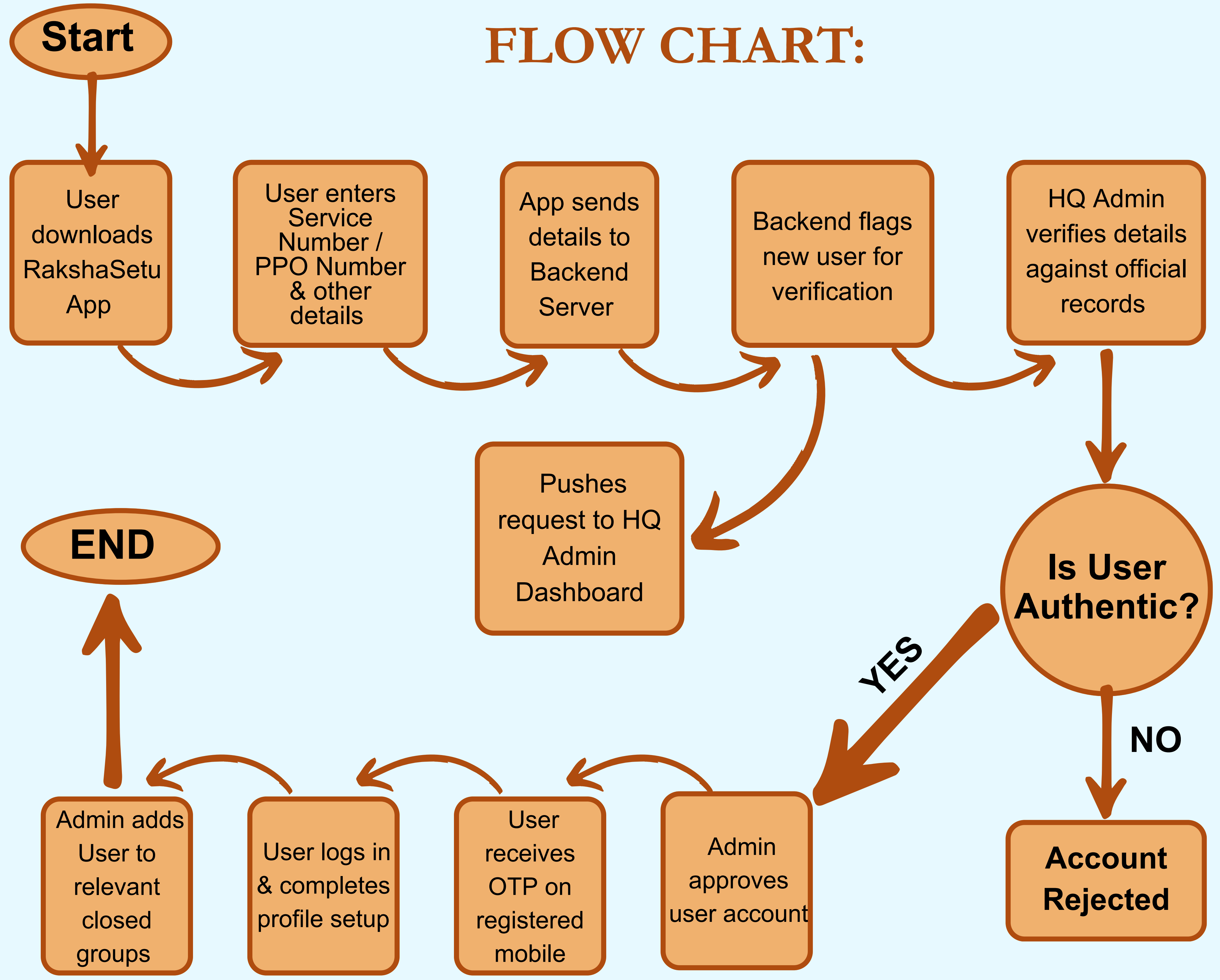
Nationwide launch with continuous monitoring, training, and updates.
Full Rollout & Maintenance

FLOW CHART:



TECH STACK

Flutter · React · node js · TypeScript · Signal · WireGuard · WebSocket · PostgreSQL · WebRTC

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FEASIBILITY AND VIABILITY

ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY

Robust encryption (Signal Protocol), VPN encapsulation (WireGuard), and secure on-premise infrastructure are achievable using mature technologies and modern backend frameworks.

2. ECONOMICAL FEASIBILITY

While initial setup requires high investment in secure infrastructure, long-term costs are manageable through controlled scaling, microservices, and military-operated servers, ensuring high value for national security.

3. SOCIAL FEASIBILITY

The platform builds trust among defence personnel, veterans, and families by ensuring safe communication, strengthening morale and reducing risks of information leakage.

4. LEGAL & COMPLIANCE FEASIBILITY

All data is hosted within Indian territory, ensuring compliance with data sovereignty, CERT-In, and government cybersecurity frameworks, making it fully legitimate and acceptable.

5. OPERATIONAL FEASIBILITY

A user-friendly mobile app with guided onboarding ensures easy adoption, while centralized HQ-admin controls simplify group management, verification, and monitoring.

6. SECURITY FEASIBILITY

Continuous penetration testing, DevSecOps pipeline, and strict DLP measures ensure resilience against cyberattacks, insider threats, and device-level compromises.

POTENTIAL CHALLENGES AND RISKS

SECURITY VULNERABILITIES

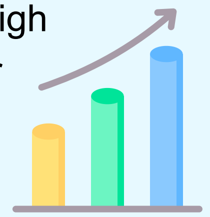
Risk of cyberattacks from state-sponsored actors targeting servers, VPN, or the app.

USER ADOPTION & TRAINING

Defence personnel and families may resist shifting from familiar apps to a new platform.

PERFORMANCE & SCALABILITY

Ensuring smooth operation under high load, especially in low-bandwidth or remote areas.



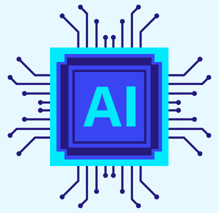
INSIDER THREATS

Malicious actions from authorized users with legitimate access.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

SECURITY HARDENING

Regular penetration testing, DevSecOps pipeline, MFA integration, and third-party security audits.



AWARENESS & TRAINING

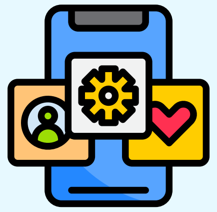
Simple UI/UX design, training modules, and campaigns to highlight risks of commercial apps.

OPTIMIZED ARCHITECTURE

Microservices design, load balancing, efficient VPN protocols like WireGuard, and offline-first features.

ACCESS CONTROL & MONITORING

Enforce least privilege principle, maintain immutable audit logs, and enable real-time anomaly detection.



IMPACT AND BENEFITS

UNIQUE FEATURES:

Mandatory VPN Encapsulation

All traffic passes through a defence-controlled VPN.

HQ-Centric Trust Model

Users and groups are verified and managed only by Defence HQ.

Total Information Containment

No data leaves the secure ecosystem.

Defence Community Exclusivity

Accessible only to personnel, veterans, and families.

Data Sovereignty Guarantee

All infrastructure hosted within India under Government Authority.

IMPACTS

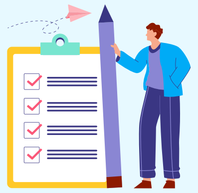
Enhanced National Security

Closes a major vulnerability by preventing hostile intelligence from exploiting communication channels.

Improved OPSEC

Reduces accidental leakage of troop movements, locations, and routines.

Family Protection

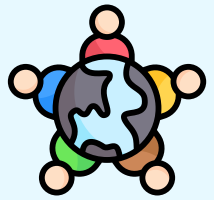


Shields families from being targeted for sensitive information or coercion.

Digital Sovereignty

Eliminates reliance on foreign platforms, strengthening India’s cyber independence.

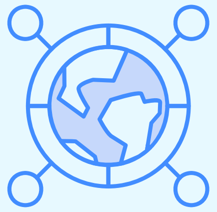
Resilient Defence Communication



Builds a secure backbone for both personal and operational communication.

BENEFITS

Trustworthy Ecosystem



Provides a safe digital environment exclusively for the defence community.

High Morale

Boosts confidence among personnel knowing their communication is secure.

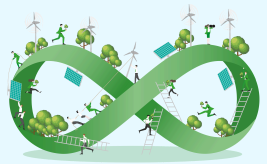
Efficient Monitoring

HQ-admin tools simplify verification, group control, and auditing.

Scalable Solution

Architecture supports growth from small groups to thousands of users.

Long-Term Cost Value



Though high setup cost, it ensures lasting security worth the investment.

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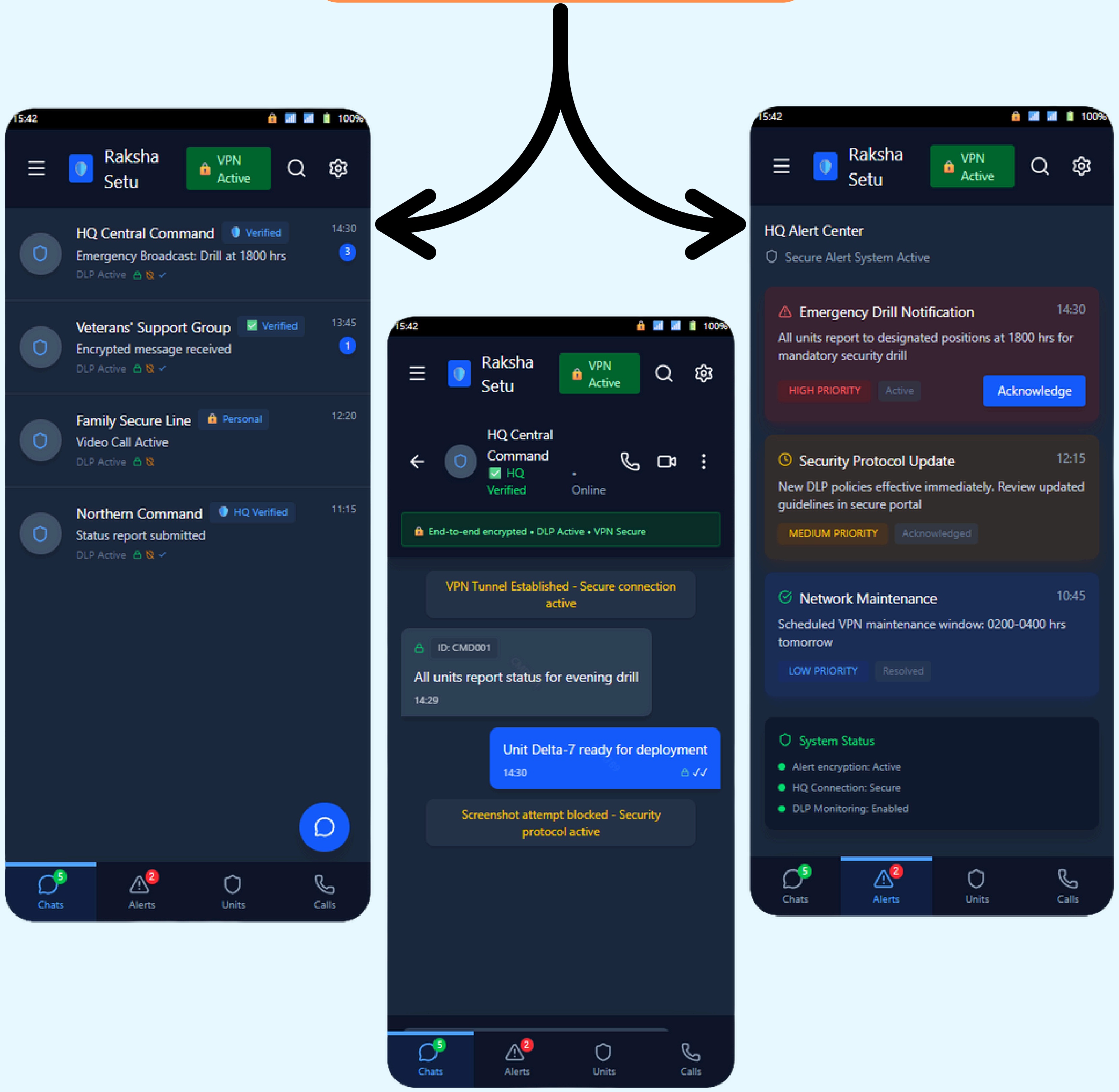
OUR PROTOTYPE

App Interface



Chat Interface

RESEARCH AND REFERENCES

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- ## Cryptographic Protocols for Secure Messaging

Summary: Explains the Signal Protocol, used in apps like Signal and WhatsApp, providing end-to-end encryption, forward secrecy, and resistance to surveillance.

Link: https://signal.org/docs/

- ## VPN Security & Performance Evaluation

Summary: Introduces WireGuard, a modern VPN protocol with superior speed, security, and simplicity compared to IPsec and OpenVPN, making it ideal for sensitive communication.

Link: https://www.wireguard.com/papers/wireguard.pdf

- ## Mobile Security & Data Leak Prevention

Summary: Defines major mobile app vulnerabilities and provides best practices for secure coding, DLP (screenshot blocking, copy-paste prevention), and device-level protection.

Link: https://owasp.org/www-project-mobile-top-10/

- ## Risks of Using Commercial Apps for Defense Communication

Summary: Using commercial apps for defense communication risks data leaks and unauthorized access due to weak encryption and controls.

Link: https://owasp.org/www-project-mobile-top-10/